The mortgage book without the deal models — 858 of 882 priced

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1. The short version

Two weeks ago this report ended on a question: *if the repayment history comes back, we will know within days whether these securities can be valued without buying deal models.*

The history came back, and the answer is yes.

- Of the 882 government mortgage securities, **505 priced a fortnight ago and 858 price now.**
- Of the **377** we listed then as unpriceable — each one named — **353 now price.** The remaining **24** are still named, and their reasons are in §4.4.
- It works across **five different kinds of security** — whole pools, slices of a deal, principal-only strips, interest-only strips and forward contracts — with **one** method and no deal rules of any kind.

⚠ **One thing we did differently from the call, and it should not be buried.** You said to treat these as bullet bonds first — all the principal returning on the legal maturity date. We amortised them instead. On the securities whose repayment history we hold, the legal maturity is a median of **24.3 years** away while the money actually came back in a median of **3.0** — about **seven times** too long, which would have added roughly twenty years of duration to more than half the book by count. ☆ And your own Tier 1 slide permits the choice: it asks for cash flows that do not respond to prepayment or to rates, and a *recorded* repayment history is exactly that — it is history, so it responds to nothing. The amortising treatment costs nothing and is the more realistic effective-life representation your Tier 2 slide already lists.

☆ **And outside the mortgage book**, the coupon formulas you sent on Tuesday let **four corporate bonds price that have been held at the custodian's mark since July**, and confirmed eight margins we had found ourselves, to the basis point (§5.2).

What is still missing is one category — the 412 non-government CMOs, commercial mortgage-backeds and asset-backeds, which have no repayment history at all. The same method will work on them; it needs the same data (§6).

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2. How it works

Picture the pool as a **bucket** that fills with money each month as a thousand borrowers make their payments. What a holding actually is comes in two shapes:

- **a percentage of the bucket.** Whatever goes in, you get your share of it.
- **one of a row of glasses the bucket is poured into, in a fixed order.** The first glass fills before the second gets a drop. ⚠ If you own the third glass, knowing how much went into the bucket tells you nothing at all until you also know the pouring order and the size of the glasses ahead of you.

⭐ **That pouring rulebook is what deal models are sold for** — and it is why two thirds of this book looked unpriceable in July.

The way out, as last report set it out, is that somebody already records what ended up in each glass. The **factor** is the share of a security's *own* original balance still outstanding — so for a glass, it is that glass's level, after the pouring has happened. Two consecutive factors give the month's cash flow:

- **principal paid** = original size × (last month's factor − this month's factor)
- **interest paid** = original size × last month's factor × rate ÷ 12

It works, and here it is working

Take `31402CZR6`, a Fannie Mae pool at 4.5% maturing 2019, which you hold at 103.57. At the valuation date its factor is **0.44041654** — 44% of the original loans are still outstanding. The first six months:

month	factor	principal returned per 100 of original	interest paid per 100 of original
1	0.42939682	1.101972	0.165156
2	0.41997097	0.942585	0.161024
3	0.41186327	0.810770	0.157489
4	0.40355866	0.830461	0.154449
5	0.39635314	0.720552	0.151334
6	0.38777872	0.857442	0.148632

The first line is `100 × (0.44041654 − 0.42939682) = 1.101972` of principal and `100 × 0.44041654 × 4.5% ÷ 12 = 0.165156` of interest. ⚠️ The interest uses *last* month's factor, because you earn on the balance that was outstanding, not the one left after the payment.

⭐ **And the check that makes it trustworthy:** adding up the principal across the whole recorded history gives **44.041654**, against a starting factor of 0.44041654 — the same number. Every dollar of original face is accounted for and none is invented. Across all 374 securities the worst discrepancy is **one part in a hundred trillion**.

One method, five shapes of security

This is the part that matters for the purchase decision. **The same two lines above handle all of these, with no special case for any of them:**

the holding	why nothing special is needed
a whole pool	the factor is the pool's own balance
a slice of a deal — 184 of them	the factor is *that slice's* balance, so the pouring has already happened before we see the number
principal only — 76	there is no interest line. Its rate is literally 0.000%
interest only — 74	there is no principal line; interest is earned on a shrinking amount
a slice that takes its interest as more debt rather than in cash	the principal line comes out negative and cancels the interest exactly — which is what such a slice pays that month: nothing

☆ **We are not reproducing the pouring rules. We are reading the glass after the pouring has finished.** That is the whole of it, and it is why no deal model is needed.

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3. What came out, and how we checked it before trusting it

858 of the 882 now carry a price, a spread and a risk number. There are two ways to know how much money comes back each month, and the report says on every row which one was used:

- **predict it** — assume a rate at which borrowers repay early, then work out the schedule. **458 securities.** ⚠ This only works for *a percentage of the bucket*, and indeed all 458 are exactly that. These are the pools already priced in September; their history was never requested, because they did not need it.
- **read the record** — the factor method above. **373 securities.** ☆ **353 of these are glasses**, where prediction cannot work however good the assumptions, because the pouring rules decide the answer and we do not have them. Reading the glass sidesteps the rules completely.
- and **27 forward contracts**, priced as forwards (§4.2).

☆ **Twenty securities were put deliberately in both groups** — ordinary pools we already priced — so the new method could be checked against numbers we already trusted before it was pointed at a single glass.

The two checks, and they are what makes the number believable

The two routes agree. Over two different populations and two entirely different ways of producing a cash flow, the spread comes out at a median **261.3 basis points** from the recorded history (223 securities) against **250.9** from the modelled schedule (458). Ten basis points apart.

And the risk numbers agree with yours. On the securities valued from their recorded history, our effective duration has a median of **2.62 years** against the custodian's own **2.45** — 0.17 years apart, from two sources that share no inputs.

⚠ For the twenty securities in both groups, the comparison is closer still, and six of them were excluded from it and said so: they had prepaid at 57–74% a year, outside the range the modelled schedule can represent, so the "model" figure for those six was just the edge of the range wearing a different label.

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4. Where Tier 1 stops — each one named, and what closes it

4.1 Interest-only strips: a price, and no risk number

An interest-only strip receives no principal at all — it earns interest on a balance it will never be paid. So its rate sensitivity is almost entirely **prepayment response**: when rates rise, refinancing slows, that balance survives longer, *more* interest arrives, and the security is worth **more**. Fixed cash flows cannot express that. The result is not noise, it is the wrong sign: all **74** come out with a positive duration while **49** of your custodian's durations for the same securities are negative.

⭐ **So they ship with a price and market value — exact, because they are matched to the custodian's own mark — and with the duration withheld.** Of the 74, exactly **0** carry a published duration. Your custodian's figure sits in its own column beside them as evidence; we do not put their number in our column, because a column that sometimes holds one and sometimes the other is worse than a blank.

⭐ **A sentence for your "what Phase 1 does not capture" slide, if it helps:** *an interest-only strip's rate sensitivity is almost entirely prepayment response, which fixed cash flows cannot express, so Phase 1 publishes its price and withholds its duration.*

4.2 Forward contracts: two different sensitivities, not one

Twenty-seven holdings are agreements to buy an unspecified pool on a date weeks after the valuation date. Everywhere else in this book a move in the base rate and a move in the spread are the same arithmetic — across the other **757** securities the two sensitivities differ on **none**. On a forward they differ on **all 27**, by exactly the settlement period, and the reason is plain: **a rate move changes both what the pool is worth and what it costs to finance it until settlement; a spread move changes only what it is worth.**

⚠ So on those 27 rows the two are not interchangeable and are reported separately. Their duration also runs longer than your custodian's — median **2.85 years** against **1.82**, 5 of the 27 more than 1.5 years apart. That gap is the prepayment option Tier 1 does not model, so both numbers are published side by side rather than one withheld.

4.3 The 132 securities whose coupon moves

Your coupon file (§5.1) told us something the holdings file cannot: whether each security's coupon is fixed. **192 of the 882 are not fixed-rate**, and **132 of those carry a published duration**.

⭐ **The valuations are unaffected** — the coupon level we use is the custodian's own at the valuation date. What a fixed-coupon duration assumes is not the level but that the level *stays*. Those 132 now say so on the row. Where we can see the principal repaying month by month, only the interest moves with the coupon — about a sixth of the value — so the error is bounded at roughly **1.8 years**, against the twenty that a bullet treatment would have cost. ⚠ For 61 of the 132 we have no recorded principal either, so that bound does not apply to them and each of those rows says so.

4.4 The 24 that still do not price

reason	n
neither a recorded repayment history nor usable pool terms	19
the balance is already zero at the valuation date — nothing left to pay	2
the forward's description does not state a settlement month	2
the recorded history produces no net cash at all	1

⚠️ None of these is a shortfall against the plan; three of them could only be discovered by reading the history, and the other 21 are named data gaps rather than failures.

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5. What the coupon files bought — and the one thing they cannot do

5.1 The bit the holdings file cannot carry

The custodian reports each security's coupon *rate*. It never says whether that rate **moves**, and that single fact decides whether a duration is near zero or a full remaining life. Your file answers it for all **1,333** securitised and agency holdings.

⭐ It also replaced guesswork rather than refining it: we had been inferring which coupons move from the security descriptions, and scored against your answer that inference was only **40%** right — a private-label deal name simply does not say.

5.2 Four corporate bonds that had been stuck since July

The formulas include the corporate book, and four holdings that have sat at the custodian's mark since July — because no public source states what their coupon becomes after the call date — now price:

	price	implied spread
Chuo Mitsui perpetual 5.506%	42.0	**1,102.5 bp**
Resona perpetual 5.85%	51.0	**919.6 bp**
Resona euro perpetual 4.125%	40.5	**1,100.7 bp**
...and its sister tranche	39.5	**1,132.4 bp**

About where distressed Japanese bank capital belonged in March 2009. ⭐ One of the four we had recorded as "all terms unknown"; it came back identical to its sister tranche, which also confirms the two are one issue in two wrappers. Eight further margins we had found ourselves from prospectuses agree with your file **to the basis point**, with the same switch dates.

5.3 ⭐⭐ What the formulas cannot do — and what that says about Tier 2

We put the formulas into the valuation code as dated rules, exactly as you described: one bond, several periods, each with its own start date, index and margin.

⚠️ **For the adjustable-rate pools they still do not produce the right coupon.** Worked out at 31 March 2009 the formula gives a rate **one to three points below** what the custodian reports — all seventeen we can currently test, every one in the same direction. The reason is the reset: these pools reset **once a year on their own anniversary**, so in March 2009 one is carrying a rate set the previous June and another one set in December, and the one-year Treasury moved about **two and a half points** across that window. The formula is right. It cannot tell us which date to apply it on.

⭐ **The conclusion is about Tier 2, not about the formulas.** For the principal leg, the empirical route turned out to be a *recorded history* rather than a model — that is §2. This fortnight showed the **coupon leg is the same shape**: what closes it is the realised monthly coupon, because that single series records what the formula, the index, the reset date and the rate caps all did, none of which we can reconstruct separately. ⚠️ Until then, the custodian's observed coupon — the actual rate after caps and resets — remains the better input, and that is a deliberate choice rather than a gap.

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6. What is still missing

The 412. The non-government CMOs, the commercial mortgage-backeds and the asset-backeds have **no repayment history of any kind**. They are **18.2% of the portfolio by number of holdings and 4.8% by par value** — the count makes them sound larger than they are, but they are the last category with nothing behind them. **201 of the 412 (49%) also have a moving coupon**, so they need both series.

⭐ One obstacle has already gone: 37 of the 412 had no usable identifier in the custodian file and we expected to have to leave them out. Your mortgage file supplied identifiers for all 37, so the request covers the full 412.

Two requests have gone across separately, kept apart because they are different decisions: a small one to finish the government book (six index histories, and either the reset dates and rate caps or the realised coupon for its 192 moving-coupon pools), and a larger one for the 412.

Appendix — the engineering, briefly

Running it. `PYTHONPATH=src python scripts/tier1_structured.py` writes `outputs/tier1_structured_2009-03-31.csv`, 882 rows, alongside a companion file accounting for every security that did not price. The pool and corporate books have their own commands; all outputs and their checksums are listed in `docs/release_facts_2026-10-15.md` — the file numbered `07_release_facts` in the folder you received — which is generated from the files rather than typed.

Nothing else moved. Every other output is unchanged from the 2026-10-08 record except the corporate book, which gained the four bonds in §5.2 — verified row by row, with the row count, the column list and every other value identical. The test suite stands at **753**.

One internal note. The prepayment speed assumed by the 458 predicted rows was measured as a *total fall in balance* rather than a *prepayment rate* — the two differ by the contractual repayment inside every monthly payment. It is right for this population's typical remaining term and no security's duration moves by the 1.5-year threshold, but it understates the spread of the 46 shortest pools by about 47 basis points. The proper fix is a speed derived per pool, which is a prepayment model, which is Tier 2.

Terminology. The spread reported throughout is an **implied, bond-equivalent spread**, in your own words from the methodology deck — not an option-adjusted spread, which would require modelling the prepayment option explicitly. Nothing in the output is labelled otherwise.